Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/23/2021
PI Name:	Tewari, Surendra Ph.D.		
Project Title:	Spurious Grain Formation in Alloys Directionally Solidified on the Space Station Due to Marangoni Convection		
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Program/Discipline:			
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Contact Monitor:	Panda, Binayak	**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	Purpose of MICAST (Microstructure Formation in Casting of Technical Alloys under Diffusive and Magnetically Controlled Convective Conditions) experiments was to directionally solidify (DS) primary dendrite arrays along direction in an Al-7 wt% Si alloy on International Space Station (ISS) in the absence of convection and compare their spacing with those solidified on Earth (presence of convection). Three terrestrially grown, oriented, dendritic feed-stock cylindrical samples were partially re-melted and then DSed on the Space Station (speed ranging from 5 to 50 microm/s: MICAST-6, MICAST-7, and MICST-12). The observed primary dendrite spacing showed good agreement with predictions from a theoretical model which assumes diffusive transport and neglects convection. The radial macrosegregation caused by thermosolutal convection in Earth-grown samples is not seen in the MICAST samples, further confirming their convection free DS. However, a comparison of the transverse microstructures from the portion solidified on the Space Station with that from the unmelted seed shows new grains, having orientations very different from that in the oriented feed-stock, forming during DS in space. Bulk undercooling of melt (which can cause independent nucleation) will not occur during DS of alloys under positive thermal gradient in a constant diameter mold. Thermosolutal convection will also not occur in micro-g. Even if the side-arms detached from the dendrites, for example during partial remelting of the feed rods, they would maintain their orientation and position in the surrounding liquid, as shown by Glicksman during micro-g dendrite re-melting experiments. Then, how did the spurious grains form in MICAST samples? Observation of surface voids in MICAST samples raised an intriguing possibility. Is Marangoni convection caused by such surface voids (free liquid-vapor surface under a thermal gradient) responsible for the formation of spurious grains during DS in micro-g? Examination of PFMI (Pore Formation and Mobility Investigation)-videos where Succinonitrile-0.5%H2O having dissolved nitrogen was DSed on the Space Station further supports this hypothesis. We see fluid recirculation in the vicinity of surface gas pore in one PFMI experiment. We see side-branch of primary dendrites in the vicinity of a surface pore appearing to break-away and rotate, as if falling sideways, in another PFMI experiment! The mechanism of spurious grain formation due to surface-void driven Marangoni convection has not been studied. Presence of spurious grains is highly detrimental to the mechanical properties, especially high temperature creep and thermal fatigue. Purpose of this research is to develop a three dimensional solidification model incorporating Marangoni convection and thermosolutal convection which can predict formation of spurious grains during DS. We intend to extract data, such as, fluid velocities along stream lines as a function of distance from the surface-void, angular rotation of the fragmented side arms due to Marangoni flow lines, etc. from PFMI (Pore Formation and Mobility Investigation) videos, and then compare these observations with the simulations in order to quantitatively validate our numerical model. We will then use this model to predict formation of spurious grains in a totally different alloy, Al-7Si (MICAST), under varying processing conditions, such as, during re-melting prior to DS, after a step increase or decrease in growth speed vs. at constant speed, etc. Microstructures of MICAST samples in the vicinity of surface pores will be used to compare simulation predictions with the observed propensity of nucleation and growth of spurious grains in micro-g. Physical properties required for simulations are available for both alloys. Results from this research will be highly valuable not only for terrestrial technologies involving solidification, such as repairing gas turbine blade tips by laser cladding, but also for any future space manufacture or repair which involves solidification. Reference: M. E. Glicksman, A. Lupulescu, M.B. Koss: Melting process under microgravity conditions, Adv. Space Res., 32 (2003) 237-242.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will be highly valuable not only for terrestrial technologies involving solidification, such as repairing gas turbine blade tips by laser cladding, but also for any future space manufacture or repair which involves solidification.
Task Progress:	Presented below is the summary of the two research tasks performed under this project so far: 1. Dendritic Solidification of Succinonitrile-0.24 wt% Water Alloy: A Comparison with Microgravity Experiments for Validating Dendrite Tip Velocity Summary The Pore Formation and Mobility Investigation (PFMI) at the International Space Station provided information on the morphological evolution during remelting and directional solidification under microgravity conditions for Succinonitrile-0.24 wt% water binary alloys. Unlike the terrestrial experiments where the growth is affected by natural convection, constrained diffusive growth is expected in the microgravity experiments. This study aimed at providing an experimental benchmark of dendritic growth applicable for validation of theoretical and numerical dendrite growth models using the PFMI video data. Succinonitrile-water dendrite arrays grew out, under diffusive conditions, from both sides of a liquid zone generated within the initially directionally solidified sample. The temperature distribution and tip velocity of growing dendrites were measured from the PFMI videos during directional solidification of these growing dendrites. Columnar dendrite growth was simulated using phase field and cellular automata models in two and three dimensions, while the lattice Boltzmann method was employed to solve for the solute transport equation. The tip velocities predicted by both models were validated with the Lipton, Glicksman, and Kurz analytical model. Then, the columnar growth of the Succinonitrile-water alloy was simulated with initial boundary conditions extracted from the PFMI experiment. Both cellular automata and phase field models were capable of predicting the average tip growth in agreement with the observed behavior. However, the computational time was almost ten times larger for the phase field simulations. Some differences were observed in the dendritic morphologies predicted by the cellular automata and phase field models, which can be attributed to the way in which each method treats the interface. 2. Origin of Spurious Grain Formation during Directional Solidification of Alloys in micro-g Environment of International Space Station. Summary During directional solidification of alloys in the microgravity environment of the International Space Station (ISS), growth of dendritic array is expected to be under purely diffusive transport conditions. The resulting microstructure is

	expected to be made up of uniformly arranged primary dendrites on sample cross-sections without any defects, such as macrosegregation and spurious (misoriented) grains. However, spurious grains have been seen in Al-7 wt%Si samples recently directionally solidified on the Space Station under the joint NASA-ESA (European Space Agency) project (MICAST). Spurious grains have also been observed in SCN-0.24 wt% H2O samples grown under the NASA-PFMI project. Careful examination of both of these experiments shows the presence of voids at the melt-crucible interface which are the likely the source of the convective flows responsible for fragmentation of the dendrite-side arm, and its rotation which creates the nucleus for the subsequent growth and formation of spurious grains. In this paper, we assume that side-arm of a primary dendrite has gotten detached from the primary trunk during re-melting and isothermal hold, prior to the onset of directional solidification on the Space Station, in the vicinity of a pore at the melt-crucible interface. We numerically simulate the influence of the Marangoni convection on the trajectory of such a dendrite fragment in the alloy melt and compare with the experimental observations. The experimentally observed rotation behavior of the fragmented side-arm in transparent SCN-0.24 wt% H2O, observed from the PFMI-video, shows an excellent quantitative agreement with simulation results. The strength of Marangoni convection is predicted to be much higher in the presence of surface voids seen in the Al-7 wt% Si MICAST samples. The predicted rotational speeds of broken-off secondary arm in the Al-7 wt% Si (MICAST) samples is significantly higher than those in the SCN-0.24 wt% H2O (PFMI). The strength of the convection is dependent on the Marangoni Number and location of the surface pore relative to the Mushy zone. Marangoni convection is strong enough to force the rotation of broken-off side arms and should be considered an important source of microstructural inhomogeneity during terrestrial applications, such as laser additive metal deposition.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Nabavizadeh SA, Lenart R, Eshraghi M, Felicelli SD, Tewari SN, Grugel RN. "Dendritic solidification of Succinonitrile-0.24 wt% water alloy: A comparison with microgravity experiments for validating dendrite tip velocity." Acta Astronautica. 2020 Oct;175:163-73. https://doi.org/10.1016/j.actaastro.2020.05.059 , Oct-2020
Articles in Peer-reviewed Journals	Masoud G, Upadhyay SR, Rajamure RS, Tewari SN, Grugel RN, Poirier DR, Lauer M. "Primary dendrite array morphology in Al-7 wt% Si alloy samples directionally solidified aboard the International Space Station." J. Cryst. Growth. 2021 May;562:126077. https://doi.org/10.1016/j.jcrysgro.2021.126077 , May-2021